PAPER #91 ■ MUGE Resonance: 14-Mode Gravity Framework

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Title: MUGE Resonance Gravity: 14-Mode Framework from aDPM Base Through Wormhole Metric

Author: Daniel T. Murphy **Framework:** MUGE Resonance, UQFF Star-Magic ([SSq] = 0.57, [SCm] ■ 0.99) **Date:** March 7, 2026 **Source Data:** validate*uqff*muge.py, source4.cpp (14 Resonance functions), compute*resonance*MUGE_SOURCE4 **Index Slot:** ■1.12 UQFF Master Calculators, Paper #91

Abstract

MUGE Resonance extends compressed gravity with 14 mode-specific corrections, beginning from the anomalous Doppler modulation (aDPM) base and including terahertz, vacuum diffusion, super-frequency, aether resonance, Ug4 intensity, quantum frequency, aether frequency, fluid frequency, oscillation, expansion frequency, Toroidal Resonance Zone, and wormhole metric contributions. The `compute_resonance_MUGE_SOURCE4` function returns a complete resonance gravity value for any astrophysical system; `validate_uqff_muge.py` confirms all 14 terms are finite across 5 systems.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. The 14-Mode Resonance Decomposition

MUGE Resonance gravity:

```
g_{\rm MUGE}^{\rm Res}(r, \vec{\omega}) = g_{\rm aDPM}(r) + \sum_{k=1}^{13} \delta_k^{\rm Res}(r, \vec{\omega})
```

Mode k	Symbol	Frequency/Origin	Physical Effect
Base	aDPM	Anomalous Doppler	Doppler-modified gravity
1	aTHz	THz frequency	Terahertz coupling
2	Avac_diff	Vacuum diffusion	Vacuum polarization diffusion
3	aSuperFreq	SuperFreq (SGR1745)	Magnetar super-frequency
4	aAetherRes	Aether resonance	Quantum vacuum resonance
5	Ug4_i	Ug4 intensity mode	BH vacuum concentration
6	aQuantumFreq	QuantumFreq	Quantum oscillation modes
7	aAetherFreq	AetherFreq	Aether field frequency
8	aFluidFreq	FluidFreq	Navier-Stokes fluid resonance
9	Osc_term	General oscillation	Composite oscillation
10	aExpFreq	ExpFreq	Hubble expansion frequency

Mode k	Symbol	Frequency/Origin	Physical Effect
11	fTRZ	Toroidal Resonance Zone	f_TRZ = 0.01
12	Wormhole	Wormhole metric	Einstein-Rosen bridge topology

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2. aDPM Base Grace

The anomalous Doppler modulation base:

```
g_{\rm aDPM}(r, v) = \frac{GM}{r^2} \cdot \frac{1 - v/c}{1 + v/c}
```

For bound circular orbital: v = $(GM/r)^{1/2}$, giving:

```
g_{\rm aDPM}(r) = g_{\rm Newton}(r) \cdot \left(1 - 2\sqrt{R_S/r}\right)^{1/2}
```

? At r = 10 R_S: correction = -6.3■0.1% (sub-GR post-Newtonian)

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3. 5-Frequency Resonance (Source27/28 Origin)

Modes 1,3,6,7,8 correspond to the **5-frequency SuperFreq resonance** from source27/28:

Frequency	Symbol	Origin System
SuperFreq	aSuperFreq	SGR1745 Magnetar
QuantumFreq	aQuantumFreq	SgrA*
AetherFreq	aAetherFreq	Universal vacuum
FluidFreq	aFluidFreq	Accretion disk
ExpFreq	aExpFreq	Hubble expansion

Combined 5-frequency resonance amplitude:

```
A_{\rm 5freq} = \prod_{k} \left(1 + a_k \cos(\omega_k t)\right) \approx 1 + \sum_k a_k
\cos(\omega_k t)
```

For small amplitudes `a_k \ll 1`.

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4. TRZ Mode (fTRZ Factor)

The Toroidal Resonance Zone mode:

```
\delta_{\rm TRZ}(r) = f_{\rm TRZ} \cdot g_{\rm aDPM}(r) = 0.01 \times g_{\rm aDPM}(r)
```

This is the same f*TRZ = 0.01 that modifies Hawking temperature (Paper #81) ■ a universal UQFF factor. At the horizon scale, d*TRZ = 0.01 ■ ?g provides a 1% enhancement observable in precision pulsar timing.

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5. Wormhole Metric Contribution

The wormhole mode uses Ellis drainhole metric:

```
\delta_{\rm WH}(r) = g_{\rm aDPM}(r) \cdot \exp\left(-\frac{r^2}{l_{\rm WH}^2}\right)
```

Where l*WH = Planck-scale wormhole throat. For r >> l*WH: d*WH ■ *0 (undetectable). For Planck-regime:* *d*WH ■ g_aDPM (full wormhole topology correction).

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6. Validation Results (validate*uqff*muge.py)

All 14 resonance modes computed for all 5 systems:

System	g_total^Res (m/s■)	All modes finite	aDPM correction
Sgr A* (r_horizon)	238.4	?	-1.7%
M87* (r_horizon)	2261	?	-1.9%
Sun (surface)	273.8	?	-0.003%
NeutronStar (surface)	1.63■10■■	?	-5.1%
Magnetar (surface)	1.75■10■■	?	-5.2%

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7. Comparison: Compressed vs Resonance

Property	MUGE Compressed	MUGE Resonance
Terms	10 (static corrections)	14 (frequency-dependent)
Primary physics	Multi-scale corrections	Oscillation modes
Stable results	Always	When ?_k bounded
Dominant regime	Galaxy■cosmological	Near-compact object
TRZ included	No (Compressed uses d_Ug4 only)	Yes (explicit fTRZ mode)
Wormhole	No	Yes (Planck-scale)

Summary

The MUGE Resonance 14-mode framework provides the most complete gravity description for compact object environments, combining the 5-frequency resonance from source27/28 with TRZ, aDPM Doppler correction, and Planck-scale wormhole topology. All 14 modes are finite for 5 astrophysical systems.

Source: validateuqffmuge.py | source4.cpp computeresonanceMUGE_SOURCE4 | 14 modes ■ 5 systems all finite

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See also: PAPER_090 | Part of the Star-Magic UQFF Whitepaper Series.

UQFF computed: MUGE buoyancy ratio U*bi/F*U = [SSq]■?■r■/GM = 5.7e-1■5.0e-4 = 2.85e-4; compressed MUGE baseline g = 5.4e-7 m/s■ at r_ISCO.
